

A Multi-Source Dataset on Constituency Development Fund (CDF) Disbursements, Administrative Wards, and Public Service Facilities for Namwala Town Council, Zambia

Project Team #49, CSC 4792: Data Mining and Warehousing

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Abstract

This article describes a curated, multi-source dataset covering Namwala Town Council, a local authority in Namwala District, Southern Province, Zambia. The dataset was assembled for a data mining and warehousing course mini-project requiring extraction, cleaning and curation of council-level data from a council's official digital footprint. Following an unsuccessful attempt to scrape the council's official website directly (disallowed by the site's robots.txt at the time of collection), data were instead compiled from verified secondary and government open-data sources: national census statistics, a national geospatial public-facility registry, and news coverage quoting named council officials on Constituency Development Fund (CDF) disbursements. The dataset comprises seven pipe-separated CSV files covering the council's administrative profile, its fourteen wards and their 2010 census populations, a sample of public service facilities with geocoordinates, 2022 CDF Empowerment Grant disbursement records at quarterly and beneficiary level, the 2025/26 CDF grant-call categories and eligibility rules, and the national legal instruments governing council finance. All records carry an explicit source-provenance field.

Specifications Table

Subject	Computer Science — Data Mining and Warehousing; Public Administration / e-Government data
Specific subject area	Local government finance and administrative data for a Zambian town council
Type of data	Tables (CSV)
Data collection	Secondary-source web extraction (news articles, census tables, government open-data facility registry)
Data source location	Namwala Town Council, Namwala District, Southern Province, Zambia (15.75°S, 26.43°E)
Data accessibility	Repository name: Kaggle. Data identifier: db-unza26-csc4792-namwala-town-council-dataset. Direct
Related research article	Not applicable

Value of the Data

- The dataset provides a rare, machine-readable, council-level view of Constituency Development Fund (CDF) disbursement patterns in a small Zambian town council, an area for which structured, openly reusable data is scarce despite the CDF's growing share of local development financing.
- Researchers studying decentralisation, local governance, or public financial management in Zambia can reuse the CDF and legal-framework files to compare Namwala against other councils once equivalent datasets exist, or to track disbursement patterns over subsequent years.
- The ward-population and public-facility files support spatial and service-delivery analyses (e.g. facility-to-population ratios per ward), useful to planners, NGOs and other student groups building on the same council or replicating the pipeline for a different council.

- The explicit per-record source-provenance fields and the documented robots.txt finding make the collection methodology itself reusable as a template for scraping other .gov.zm council sites, several of which are likely to present the same access restriction.

Background

This dataset was compiled as part of the CSC 4792 (Data Mining and Warehousing) mini-project at the University of Zambia, which assigned each project team a specific City, Municipal, or Town Council and required extraction of data reflecting the functions and financial footprint of that council, as mandated under the Local Government Act No. 2 of 2019 and its 2023 and 2026 amendments. Project Team #49 was assigned Namwala Town Council. The brief specifically called for CDF allocation/disbursement/project-status data, budget and locally generated revenue data, Integrated Development Plan and community project records, and administrative and ward-level data. This article documents how those requirements were met once direct scraping of the council's own website proved infeasible.

Data Description

db-unza26-csc4792-namwala_council_profile.csv (26 rows). Long-format attribute/value table describing the council's administrative identity, location, population figures at three census points (2000, 2010, 2022), and the three governing Acts.

db-unza26-csc4792-namwala_wards.csv (14 rows). One row per administrative ward in Namwala District with its 2010 census population.

db-unza26-csc4792-namwala_public_facilities.csv (17 rows). Public service facilities (schools, health centres, a hospital, a post office) located chiefly in Namwala Central and Baambwe wards, each with a facility code, type and WGS84 latitude/longitude.

db-unza26-csc4792-namwala_cdf_empowerment_grant_2022_quarterly.csv (4 rows). Quarterly totals (amount in Zambian Kwacha, disbursement date, beneficiary count) for the 2022 CDF Empowerment Grant cycle, as reported by the council's Socio-Economic Planner.

db-unza26-csc4792-namwala_cdf_empowerment_grant_2022_beneficiaries.csv (12 rows). Named individual beneficiary groups from the 2022 CDF Empowerment Grant (a documented subsample of the 93 total beneficiaries reported), with ward, amount and project type.

db-unza26-csc4792-namwala_cdf_2025_26_grant_categories.csv (4 rows). The four CDF funding categories open for application in the 2025/26 cycle for Namwala Constituency, with grant ceilings, eligibility criteria and the application deadline.

db-unza26-csc4792-namwala_legal_framework.csv (3 rows). The three national legal instruments governing council finance referenced in the project brief, with the specific provision relevant to council financing and a citable source URL.

Methods

Study design. The collection followed the exemplar workflow demonstrated in class (Phiri, 2026): identify candidate sources, extract, clean, document provenance, and export in a standardised, separator-delimited format.

Primary-source access attempt. An HTTP request and a robots.txt check were made against <https://www.namwalacouncil.gov.zm> using Python's requests, BeautifulSoup and urllib.robotparser libraries. Automated access was disallowed at the time of collection (September 2026), so the council's official domain could not be scraped directly.

Secondary-source extraction. Four categories of secondary sources were used instead, each accessed with requests/BeautifulSoup (for HTML tables) or targeted regular expressions (for prose news content): (i) Zambia Monitor, a Zambian news outlet, for a report quoting the council's Socio-Economic Planner on 2022 CDF Empowerment Grant disbursements; (ii) Funds for NGOs, a funding-opportunities aggregator, for the council's 2025/26 CDF grant-call notice; (iii) citypopulation.de, presenting Zambia Central Statistical Office census tables, for ward and district population figures; (iv) the Smart Zambia PDU Digital Navigator, a government open geospatial facility registry, for public facility records with coordinates.

Cleaning and preprocessing. Extracted tables were standardised to snake_case column names; currency and population figures were coerced to numeric types after stripping non-numeric characters; candidate duplicate rows were checked against natural keys (ward_name, facility_code, or quarter + beneficiary_name); and no values were imputed for data points not found in an openly citable source (e.g. FY2026 locally generated revenue), which were instead omitted. Every retained row carries a source or source_url field.

Verification. All figures extracted from prose (the CDF news report) were manually cross-checked against the original article text before inclusion.

Data source location: Namwala Town Council, Namwala District, Southern Province, Zambia. **Data accessibility:** the dataset is deposited on Kaggle (db-unza26-csc4792-namwala-town-council-dataset) under a CC-BY licence consistent with the exemplar case-study dataset (Phiri, 2026); the accompanying extraction/cleaning code is version-controlled on GitHub.

Ethics Statement

All data used are publicly available administrative, statistical, and news information; no human subjects research, personal/sensitive data, or animal data were involved. Named beneficiary groups in the CDF file are cooperative/club names already published by the council and reported in the public press, not individual persons.

CRedit Authorship Contribution Statement

Project Team #49: Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. [Insert individual member names and specific CRediT roles before submission.]

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The raw and cleaned CSV files, the extraction/cleaning Jupyter notebook, and this data description paper are available at the Kaggle dataset and GitHub repository links given in the Specifications Table above.

Acknowledgments

The authors thank the course instructor for the exemplar Kaggle case study used as a reference standard for this dataset (Phiri, 2026).

References

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[2] Local Government (Amendment) Act, 2023. (2023, December 26). <https://zambialii.org/akn/zm/act/2023/28/eng@2023-12-26>

[3] Local Government (Amendment) Act, 2026. (2026, June 12). <https://zambialii.org/akn/zm/act/2026/76/eng@2026-06-12>

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